

(1) EM Spectrum

Ranges	γ	X	UV	vis	IR	micro	radio
Energy (E)	$\uparrow$						$\downarrow$
Frequency (ν)	$\uparrow$						$\downarrow$
Wavelength (λ)	$\downarrow$						$\uparrow$
Wavelength (λ)	<0.01	0.01–10	10–400	400–700	700–1e6	1e6–9	>1e9

• violet • indigo • blue • green • yellow • orange • red

(2) $c = \lambda\nu$ (Speed of light = wavelength $\times$ frequency)

c	ν (Hz)	kHz	MHz	GHz
λ (m)	2.9979e8 (m/s)	2.9979e5 (m·kHz)	2.9979e2 (m·MHz)	2.9979e-1 (m·GHz)
nm	2.9979e17 (nm·Hz)	2.9979e14 (nm·kHz)	2.9979e11 (nm·MHz)	2.9979e8 (nm·GHz)

(3) $E = h\nu = h(c/\lambda)$ (Energy = Planck $\times$ frequency)

hc , h	λ (m)	λ (nm)	ν (Hz)
E (J / photon)	1.9864e-25 (J·m)	1.9864e-16 (J·nm)	6.626e-34 (J·s)
J / mol	1.1962e-1 (J·m/mol)	1.1962e8 (J·nm/mol)	3.9902e-10 (J·s/mol)
kJ / mol	1.1962e-4 (kJ·m/mol)	1.1962e5 (kJ·nm/mol)	3.9902e-13 (kJ·s/mol)

• Grams $\rightarrow$ mol $\rightarrow \times 6.022e23$ for photon count

(4) Quantum Principles

Maxwell	Light is electromagnetic radiation
Planck	Energy emitted only in fixed amounts (quanta) (black-body radiation)
Einstein	Different light has different amounts of fixed energy (photons) (photoelectric effect)
Bohr	Electrons are also quantum (line spectra); Rydberg equation
De Broglie	Electrons are particle and wave (5)
Schrodinger	Wave equation (n , l , m_l , m_s)
Heisenberg	Position and velocity not both knowable

• $E = h\nu = h(c/\lambda) = R(1/n_f^2 - 1/n_i^2)$

• $n_i = \sqrt{1/(1/n_f^2 - hc/(R\lambda))}$; $n_f = \sqrt{1/(1/n_i^2 + hc/(R\lambda))}$

Variable	Value	Variable	Value
h	6.626e-34 J·s	h/R	3.0394e-16 s
c	2.9979e8 m/s	hc	1.9864e-25 J·m
R	2.18e-18 J	hc/R	9.1120e-8 m

(5) $\lambda = h/(m \cdot u)$ (De Broglie)

- 1 nm = 1e-9 m
- 1 e $^-$ = 9.11e-31 kg

h	u (m/s)	km/s
m (kg)	6.626e-34 (J·s)	6.626e-37 (kJ·s)
g	6.626e-31 (mJ·s)	6.626e-34 (J·s)
amu	3.9892e-7 (J·s·amu/kg)	3.9892e-10 (kJ·s·amu/kg)

(6) Electron Configuration

Symbol	Name	Value
n	Principal energy level (shell)	(1, 2, 3, ...)
l	Sublevel = subshell (shape)	(0, ..., $n - 1$)
m_l	Orbital (orientations)	(- l , ..., l)
m_s	Spin (2 per orbital)	$\pm 1/2$

	l	m_l	# orbitals	# e $^-$
s	0	0	1	2
p	1	± 1	3	6
d	2	± 2	5	10
f	3	± 3	7	14

• 1s 2s 2p 3s 3p **4s 3d** 4p 5s 4d 5p 6s 4f 5d 1. Respect **s2 p6 d10 f14**

Trend	$\rightarrow$ across period	$\downarrow$ down group
Z_{eff}	increases	$\sim$ constant
Atomic radius	decreases	increases
Ionization energy	increases	decreases
Electron affinity	more favorable	less favorable

- **Cr** = [Ar]4s¹3d⁵ • **Cu** = [Ar]4s¹3d¹⁰ (half/full d wins) • same trick one row down: **Mo**, **Ag** (and Au)
- Impossible if any subshell is over-filled (3s³, 2p⁸)
- Faster than writing Aufbau: read it off the table — s-block (gp 1–2) $\rightarrow ns$, p-block (13–18) $\rightarrow np$, d-block $\rightarrow (n-1)d$; the period number is n
- Can be a transition metal; ground state only

(7) Reading an Electron Configuration

Which element?

1. Noble-gas core $\rightarrow$ its own Z : [He]2 [Ne]10 [Ar]18 [Kr]36 [Xe]54
2. Add every superscript to it = total e $^-$ = Z (neutral atom)
3. Look up Z on the periodic table

• Cross-check: outermost subshell $\rightarrow$ period = n , block $\rightarrow$ group

Paramagnetic vs diamagnetic

1. Draw boxes for the outer subshell — **s 1 • p 3 • d 5 • f 7**
2. Fill singly, parallel spins, before pairing (Hund)
3. Count unpaired e $^-$

- Any unpaired $\rightarrow$ **paramagnetic** • all paired $\rightarrow$ **diamagnetic**
- Full subshells (s², p⁶, d¹⁰) are always paired

Excited vs ground state

- **Ground** = strict Aufbau order, no gaps
- **Excited** = an e $^-$ sits higher than Aufbau requires, leaving a gap below it; same total e $^-$ count

Valence electrons

- Count e $^-$ in the **highest** n (outer s + p) = the group number, 1A $\rightarrow$ 1 ... 8A $\rightarrow$ 8
- Everything below the highest n is **core**

(8) Ion Configuration

1. Start from the neutral atom
 2. **Anion**: add e $^-$ into the outer p until it reaches a noble-gas configuration
 3. **Cation**: remove from the **highest** n **first**
- Transition metals lose **4s before 3d**
 - Exam ions are main-group, not transition metals

(9) Quantum Numbers, Penetration & Shielding

- Shell n holds n^2 orbitals, $2n^2$ e $^-$
- **energy**: **s < p < d < f** (within shell, same n)
- **Pauli exclusion** — no two e $^-$ carry the same four quantum numbers $\rightarrow$ 2 per orbital, opposite spins
- **Penetration** — orbital reaching into the core region
- $\uparrow$ shielding $\rightleftharpoons \uparrow$ reduced Z_{eff} $\rightleftharpoons \downarrow$ penetration $\rightleftharpoons \uparrow$ **energy** (reverse also holds)
- **Shielding** — only **core** (inner) e $^-$ shield effectively; valence e $^-$ shield each other poorly

(10) Z_{eff} & Core Electrons

$$Z_{eff} \approx Z - (\text{core } e^-)$$

1. Core e $^-$ = total e $^-$ – valence e $^-$ (= the **previous noble gas**)
 2. $Z_{eff} \approx Z - \text{core } e^-$
- Same period $\rightarrow$ same core, so the **rightmost** element wins
 - Poor valence shielding is why Z_{eff} rises across a period

(11) Rank Atomic Size

- Down a group **bigger** (new shell) • across a period **smaller** (Z_{eff} pulls in)
- $\uparrow$ = **lower-left** • $\downarrow$ = **upper-right**
- Group position beats period position when both apply
- cation < neutral < anion

(12) Rank Ionization Energy

- Across a period $\uparrow$ · down a group $\downarrow$ (opposite of radius)
- $\uparrow$ = **upper-right** (He $\uparrow$ of all) · $\downarrow$ = **lower-left**
- $IE_2 > IE_1$ always
- **Big jump** appears once the first **core** e^- is removed

(13) Rank Electron Affinity

- **Most eager for an e^- = halogens, upper right** — this course writes it both as "most negative EA " and "largest positive EA "; either phrasing means the same element
- Noble gases and full/half-full subshells ≈ 0 or unfavorable
- $EA(X)$ and IE of X are the same magnitude — $F + e^-$ releases exactly what $F \rightarrow F$ costs

(14) Families & Diagonal Rule

1A	alkali metals	most reactive metals; $\uparrow$ down
2A	alkaline earth	reactive, less than 1A
7A	halogens	reactive nonmetals; $\downarrow$ down
8A	noble gases	inert — full octet

- **Diagonal rule** — an element resembles its down-right neighbor: $Li \sim Mg$ · **Be** $\sim$ **Al** · $B \sim Si$

HW-only gaps — off the exam list

Oxides — metal oxide = **basic** (K_2O , MgO , Fe_2O_3 , Cr_2O_3) · nonmetal oxide = **acidic** (P_4O_{10} , SO_2 , CO_2 , Cl_2O_7) · Al_2O_3 and BeO = **amphoteric**

Formula from charges — $1A\ 1+ \cdot 2A\ 2+ \cdot 3A\ 3+ \cdot 5A\ 3- \cdot 6A\ 2- \cdot 7A\ 1-$; criss-cross and reduce

- Same-group swap keeps the pattern: $BaS \rightarrow CaO$
- $2A + 3F_2 \rightarrow 2AF_3 \Rightarrow A$ is $3+$ (**Al**) · $3Li + Z \rightarrow Li_3Z \Rightarrow Z$ is $3-$, so Mg gives **Mg $_3Z_2$**

Bond-type ID — ionic and covalent together = a salt of a **polyatomic ion** (NH_4NO_3 , K_2SO_4); all-nonmetal = covalent only

Family reactions

- $2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$
- $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

Odds & ends

- Lanthanides = $4f$ row (**Ce**) · actinides = $5f$ (**U**)
- Diatomic: $H_2\ N_2\ O_2\ F_2\ Cl_2\ Br_2\ I_2$
- Metallic character $\uparrow$ down, $\downarrow$ across $\rightarrow$ **lower-left** wins
- Isoelectronic = matches a noble gas e^- count (K^+ , Cl^- , $Ca^{2+} = Ar$)

(15) Lattice Energy

Energy **released** when gaseous ions form the solid crystal.

$$U \propto \frac{q_1 q_2}{r_1 + r_2}$$

1. Compare charges first
 2. If charges tie, smaller ions win
- $\uparrow$ charge $\rightarrow \uparrow$ **U** (dominant) · $\uparrow$ ionic radius $\rightarrow \downarrow$ **U**

(16) Electronegativity

EN = an atom's ability to attract the electrons shared in a bond.

$$\Delta EN = |EN_A - EN_B|$$

- Read EN off the chart — $EN \uparrow$ up and to the right, F highest, noble gases excluded

(17) Bond Polarity

1. Take the absolute difference (ΔEN)
2. Classify

~ 0	nonpolar covalent
$0.4 - 1.9$	polar covalent
≥ 2.0	ionic

- **"Most polar"** $\rightarrow \Delta EN$ **alone** (largest difference wins)
- **"Strongest / shortest bond"** $\rightarrow$ **bond order FIRST**: triple $>$ double $>$ single. ΔEN only breaks ties between equal orders — a $C \equiv N$ beats a high- ΔEN $Si=O$
- Larger $\Delta EN \rightarrow$ stronger, shorter bond, but the guide flags this as *not always true*

(18) Formal Charge & Resonance

$$FC = (\text{valence } e^-) - [\text{nonbonding } e^- + \# \text{ bonds}]$$

1. FC on every atom
 2. Check ΣFC = the overall charge
 3. Pick the structure with FC 's **nearest 0**
 4. Put any -1 on the **more electronegative** atom
- Equivalent resonance forms $\rightarrow$ real bonds are the **average** (e.g. $1\frac{1}{2}$)

(19) Lewis Structures x4

1. Total valence e^- — $+1$ per negative charge, -1 per positive
2. Central atom = least electronegative, never H
3. Single bonds out to every atom
4. Fill outer octets with lone pairs (H takes 2)
5. Leftover e^- onto the central atom
6. Central below 8 $\rightarrow$ pull an outer lone pair into a double/triple bond, **unless** $Al/B/Be$
7. Check formal charges — see (18)

Four structures — one of each type:

(a) Normal octet	8 valence e^- per atom (H wants 2) — H_2O , CH_4 , NH_3 , CO_2 , CH_2Cl_2 , CH_3CH_2Cl
(b) Incomplete octet	Al, B, Be only — content below 8, do not add double bonds to fix. $BF_3/BCl_3/AlI_3$ leave 6 e^- on the center; $BeCl_2$ leaves 4
(c) Expanded octet	Central atom period 3 or below (S , P , Cl , Xe , As , I) uses empty d orbitals for >8 — $AsCl_5$, XeI_4 , SO_4^{2-}
(d) Organic	$-OH$ alcohol · $C=O$ aldehyde/ketone · $-COOH$ carboxylic acid · $-NH_2$ amine

- Every C makes 4 bonds; a $-COOH$ carbon carries one **$=O$** and one **$-O-H$** ; every O gets 2 lone pairs
- Ions go in brackets with the charge outside
- If the all-single-bond form leaves the center with a nonzero FC , add double bonds to zero it (e.g. two $S=O$ in SO_4^{2-})

(20) Photoelectric Effect ($h\nu$ - binding / work)

$$KE = h\nu - \Phi = h(c/\lambda) - \Phi = \frac{1}{2}m_e u^2$$

$$u = \sqrt{(2hc/m_e)(1/\lambda - 1/\lambda_{thr})}$$

Variable	Value
h	$6.626e-34\ J \cdot s$
m_e	$9.11e-31\ kg$
hc	$1.9864e-25\ J \cdot m$
$2hc/m_e$	$4.3609e5\ m^3/s^2$

Have	Want	Use
ν_{thr} or λ_{thr}	Φ	$\Phi = h\nu_{thr} = h(c/\lambda_{thr})$
Φ	λ_{thr} (max ej. λ)	$\lambda_{thr} = hc/\Phi$
Φ	Φ per mol	$\Phi \times N_A$ ($/1e3$ for kJ/mol)
Φ, λ	u	1. $KE = h(c/\lambda) - \Phi$ 2. $u = \sqrt{2KE/m_e}$
λ, λ_{thr}	u	2nd eq above (skips Φ and KE)

(21) Heisenberg Uncertainty

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}, \quad \Delta p = m\Delta u$$

1. $\Delta p = m \cdot \Delta u$ ($kg, m/s$)
 2. $\Delta x \geq h/(4\pi \cdot \Delta p)$
- Result is the **minimum** uncertainty
 - Rearrange the same way to solve for Δu
 - Keep everything in kg, m, s

(22) Born-Haber Cycle

$$\Delta H^\circ_f = \Delta H_{sub} + \Sigma IE + \frac{1}{2}BE + EA + U$$

1. List every step with its sign: sublimation (+), IE (+), bond dissociation (+), EA (usually $-$), U ($-$)
 2. **Halve** the bond dissociation if you need only one atom
 3. Chain: elements $\rightarrow$ gaseous atoms $\rightarrow$ gaseous ions $\rightarrow$ solid
 4. Set the chain equal to ΔH°_f
 5. Solve for the single unknown
- Use IE_2 as well when the cation is $2+$
 - U runs ions(g) $\rightarrow$ solid, so it comes out **negative**; report $|U|$ if asked for a positive lattice energy

(23) Bond Enthalpy

$$\Delta H_{rxn} = \Sigma \Delta H(\text{bonds broken}) - \Sigma \Delta H(\text{bonds formed})$$

1. List bonds broken (reactants) and formed (products), using the balanced coefficients
 2. $\Sigma BE(\text{broken})$, always positive
 3. $\Sigma BE(\text{formed})$, subtracted
 4. $\Delta H = \Sigma \text{broken} - \Sigma \text{formed}$
- Bonds unchanged on both sides cancel — skip them
 - Negative ΔH = exothermic
 - For "heat released by g of X ," convert $g \rightarrow mol$ and scale